EXERCISE 2.6

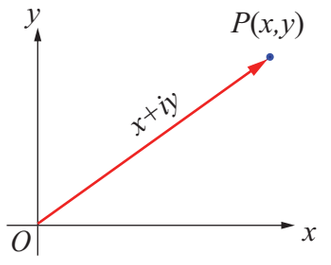
1. If $z = x + iy$ is a complex number such that $\left|\frac{z-4i}{z+4i}\right| = 1$ show that the locus of z is real axis.
2. If $z = x + iy$ is a complex number such that $\text{Im}\left(\frac{2z+1}{iz+1}\right) = 0$, show that the locus of z is $2x^2 + 2y^2 + x - 2y = 0$.
3. Obtain the Cartesian form of the locus of $z = x + iy$ in each of the following cases:
   (i) $\left[\text{Re}(iz)\right]^2 = 3$ (ii) $\text{Im}[(1-i)z + 1] = 0$ (iii) $|z+i| = |z-1|$ (iv) $\bar{z} = z^{-1}$.
4. Show that the following equations represent a circle, and, find its centre and radius.
   (i) $|z-2-i| = 3$ (ii) $|2z+2-4i| = 2$ (iii) $|3z-6+12i| = 8$.
5. Obtain the Cartesian equation for the locus of $z = x + iy$ in each of the following cases:
   (i) $|z-4| = 16$ (ii) $|z-4|^2 - |z-1|^2 = 16$.

2.7 Polar and Euler form of a Complex Number

When performing addition and subtraction of complex numbers, we use rectangular form. This is because we just add real parts and add imaginary parts; or subtract real parts, and subtract imaginary parts. When performing multiplication or finding powers or roots of complex numbers, use an alternate form namely, polar form, because it is easier to compute in polar form than in rectangular form.

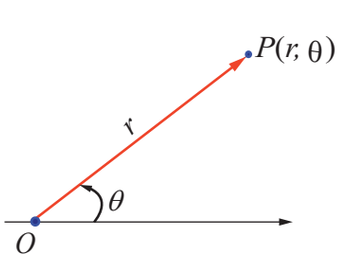
2.7.1 Polar form of a complex number

Polar coordinates form another set of parameters that characterize the vector from the origin to the point $z = x + iy$, with magnitude and direction. The polar coordinate system consists of a fixed point O called the pole and the horizontal half line emerging from the pole called the initial line (polar axis). If r is the distance from the pole to a point P and θ is an angle of inclination measured from the initial line in the counter clockwise direction to the line OP, then r and θ of the ordered pair (r, θ) are called the polar coordinates of P. Superimposing this polar coordinate system on the rectangular coordinate system, as shown in diagram, leads to



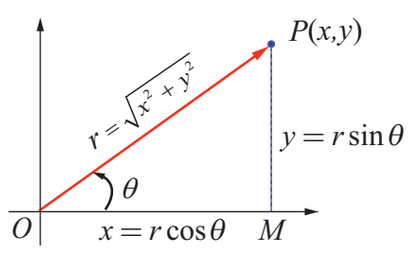
Rectangular coordinates

Fig. 2.26



Polar coordinates

Fig. 2.27



Superimpose polar coordinates on rectangular coordinates

Fig. 2.28

$$x = r\cos\theta \quad \ldots(1)$$

$$y = r\sin\theta. \quad \ldots(2)$$

Any non-zero complex number $z = x + iy$ can be expressed as $z = r\cos\theta + i\, r\sin\theta$.

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